Discrete Random Variables & Binomial Distributions

ANSWER KEY

Section 1: Expected Value of a Discrete Random Variable

1 A discrete random variable X has distribution: $P(X=0)=0.2$, $P(X=1)=0.5$, $P(X=2)=0.3$. Find $E(X)$.

a. 1.1 ✓

b. 1.0

c. 1.5

d. 0.8

$E(X)=\sum x_i P(x_i)=0(0.2)+1(0.5)+2(0.3)=0+0.5+0.6=1.1$.

2 A carnival game costs 5 dollars to play. There is a 10% chance to win 20 dollars, a 30% chance to win 5 dollars, and a 60% chance to win nothing. What is the player's expected net gain?

a. -1.50 dollars ✓

b. 3.50 dollars

c. -0.50 dollars

d. 0.00 dollars

Expected payout: $20(0.10)+5(0.30)+0(0.60)=2.00+1.50=3.50$ dollars. Net expected gain: $3.50-5.00=-1.50$ dollars.

3 If $E(X)=12$ and $Y=3X-5$, what is $E(Y)$?

a. 31 ✓

b. 36

c. 41

d. 12

By linearity of expectation: $E(Y)=E(3X-5)=3E(X)-5=3(12)-5=36-5=31$.

Section 2: Variance & Standard Deviation of Combinations

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4 Independent random variables X and Y have $\text{SD}(X) = 3$ and $\text{SD}(Y) = 4$. What is $\text{SD}(X - Y)$?

a. 5 ✓

b. 1

c. -1

d. 7

For independent variables, variances always add:
$\text{Var}(X - Y) = \text{Var}(X) + \text{Var}(Y) = 3^2 + 4^2 = 9 + 16 = 25$. Thus $\text{SD}(X - Y) = \sqrt{25} = 5$.

5 If $\text{SD}(X) = 6$, what is $\text{SD}(4X + 10)$?

a. 24 ✓

b. 34

c. 144

d. 6

Adding a constant has zero effect on spread: $\text{SD}(aX + b) = |a|\text{SD}(X) = 4(6) = 24$.

6 Why must variances be added rather than standard deviations when combining independent variables?

a. **Standard deviations are not in squared units, while the Pythagorean relationship of orthogonal independent variables governs variances** ✓

b. Standard deviations can be negative

c. Adding standard deviations always gives an imaginary number

d. Because variance is always smaller than standard deviation

By the Pythagorean property of independent random variables, variances add: $\sigma^2_{X+Y} = \sigma^2_X + \sigma^2_Y$.

Section 3: The Binomial Setting & Probability Formula

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7 Which of the following is NOT an essential requirement for a binomial setting (BINS)?

a. **The sample size must exceed 30** ✓

b. Binary outcomes (Success or Failure) for each trial

c. Independent trials

d. Fixed number of trials n, and same probability of success p

The four binomial conditions (BINS) are: Binary outcomes, Independent trials, fixed Number of trials, and Same probability of success. Sample size $n > 30$ is not a requirement.

8 A fair 6-sided die is rolled 5 times. What is the probability of rolling exactly two 4s?

a. $\binom{5}{2}\left(\frac{1}{6}\right)^2\left(\frac{5}{6}\right)^3$ ✓

b. $\left(\frac{1}{6}\right)^2\left(\frac{5}{6}\right)^3$

c. $\binom{5}{2}\left(\frac{1}{6}\right)^3\left(\frac{5}{6}\right)^2$

d. $\frac{2}{5}$

With $n = 5, k = 2, p = \frac{1}{6}$: $P(X = 2) = \binom{5}{2}\left(\frac{1}{6}\right)^2\left(\frac{5}{6}\right)^3$.

9 Evaluate $\binom{5}{2}$.

a. 10 ✓

b. 20

c. 5

d. 15

$\binom{5}{2} = \frac{5 \times 4}{2 \times 1} = 10$.

Section 4: Mean & Standard Deviation of Binomial Distributions

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10 A multiple-choice test has 20 questions, each with 4 options. If a student guesses completely at random, what are the mean and standard deviation of correct answers?

a. $\mu = 5.0$, $\sigma = 1.94$ ✓

b. $\mu = 5.0$, $\sigma = 3.75$

c. $\mu = 4.0$, $\sigma = 2.00$

d. $\mu = 5.0$, $\sigma = 15.0$

$n = 20, p = 0.25$. Mean: $\mu = np = 20(0.25) = 5.0$. Standard deviation: $\sigma = \sqrt{np(1-p)} = \sqrt{20(0.25)(0.75)} = \sqrt{3.75} \approx 1.936$.

11 When is a Normal approximation to a binomial distribution considered valid?

a. **When** $np \geq 10$ **and** $n(1-p) \geq 10$ ✓

b. When $n \geq 30$

c. When $p = 0.5$ regardless of n

d. When $np(1-p) \geq 30$

The Large Counts Condition requires expected successes $np \geq 10$ and expected failures $n(1-p) \geq 10$ for the distribution to be approximately normal.

12 In a geometric distribution where $p = 0.2$, what is the expected number of trials until the first success?

a. 5 ✓

b. 0.2

c. 4

d. 25

For a geometric random variable, the mean is $\mu = \frac{1}{p} = \frac{1}{0.2} = 5$ trials.

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